AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 5 · STUDENT LESSON

Loops and Stopping

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Before you start

Week 4: update and record a variable's state.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Trace variable values before and after each loop iteration. Try the recall before opening notes.

Your learning target

I can trace variable values before and after each **loop iteration**.

I can use the excluded stop value in range correctly.

I can explain **initialization** and test zero one and several **iteration**s.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

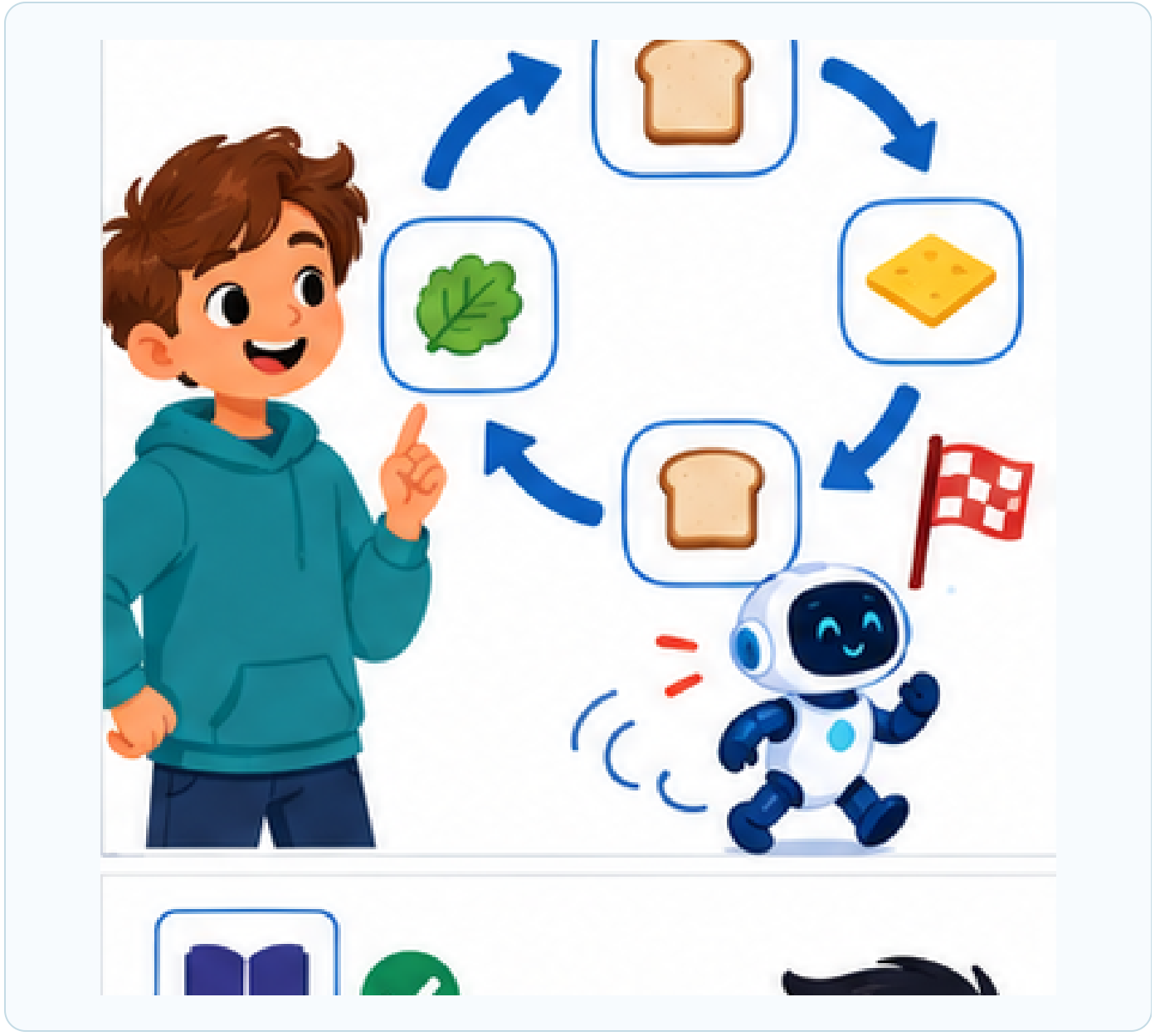
Without opening the old lesson, explain one key idea from **Trace Table**s. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Decompose a Goal. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 5: **Loop**s and Stopping

Useful repetition needs a clear **update** and **stopping rule**.



THINKER MISSION

Predict **loop** counts; repair one **infinite loop**.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

> Starting with total = 2, what does total = total + 3 store?

This week's end goal

By the end, I can trace a while-**loop** and recognise when its **update** cannot satisfy the stopping test.

STUDENT LESSON · 2 OF 18

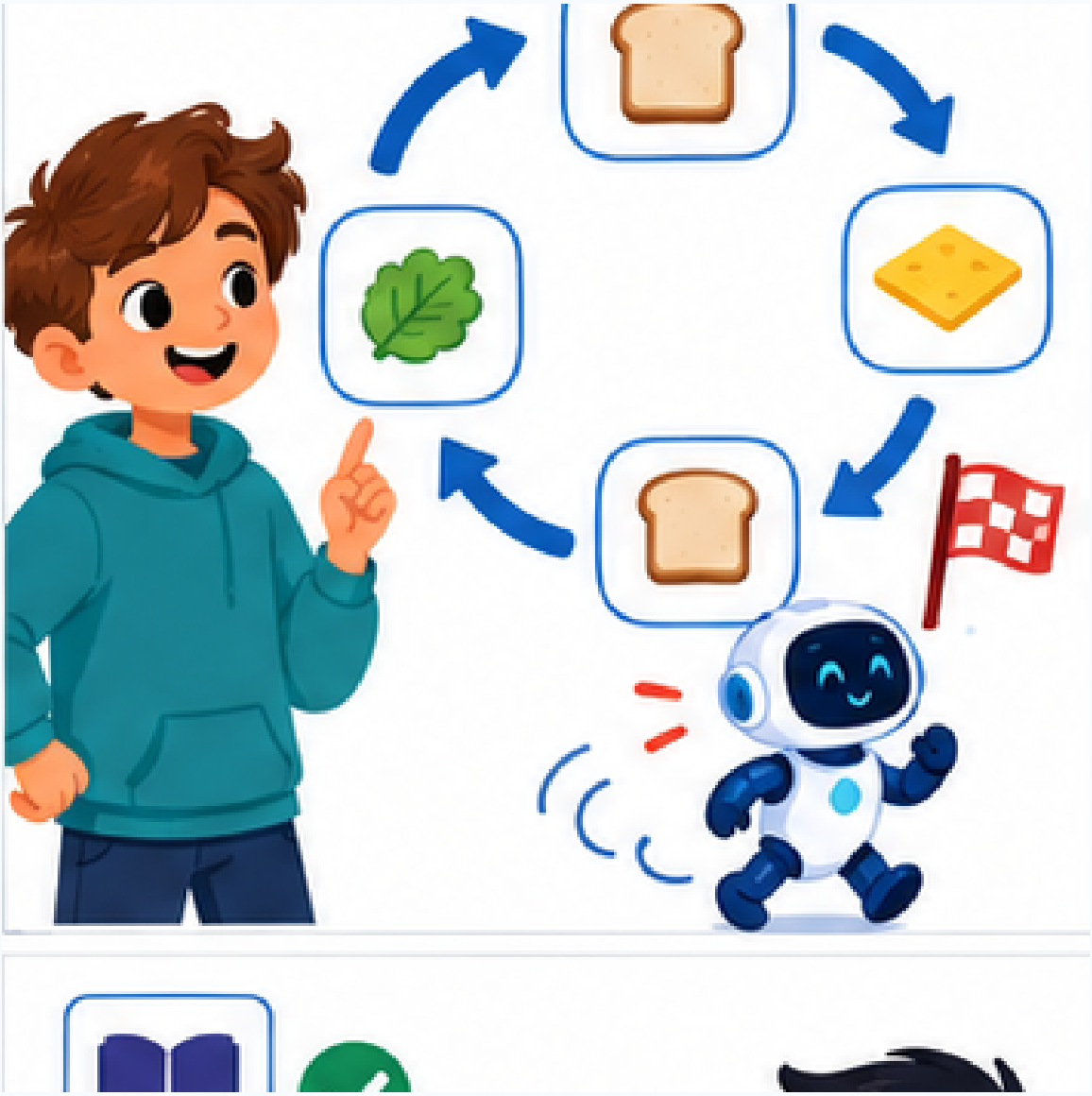
Notice the evidence

> Investigate the supplied Loops and Stopping case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use **loop**s and stopping to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

Imagine adding numbered counters: first 1, then 2, then 3. A **loop** can repeat the same addition with a different number each time. This week you will predict each change before running a short Python script.

Tracing repeated calculations prepares you to inspect the data **process**ing used in larger AI applications.

In for turn in range(1, 4), Python supplies 1, 2 and 3 in order. The start is included and the stop is excluded. The indented body runs once for each supplied value. The colon and four-space indentation mark that body.

This lesson uses integer range values with the default positive step of 1.

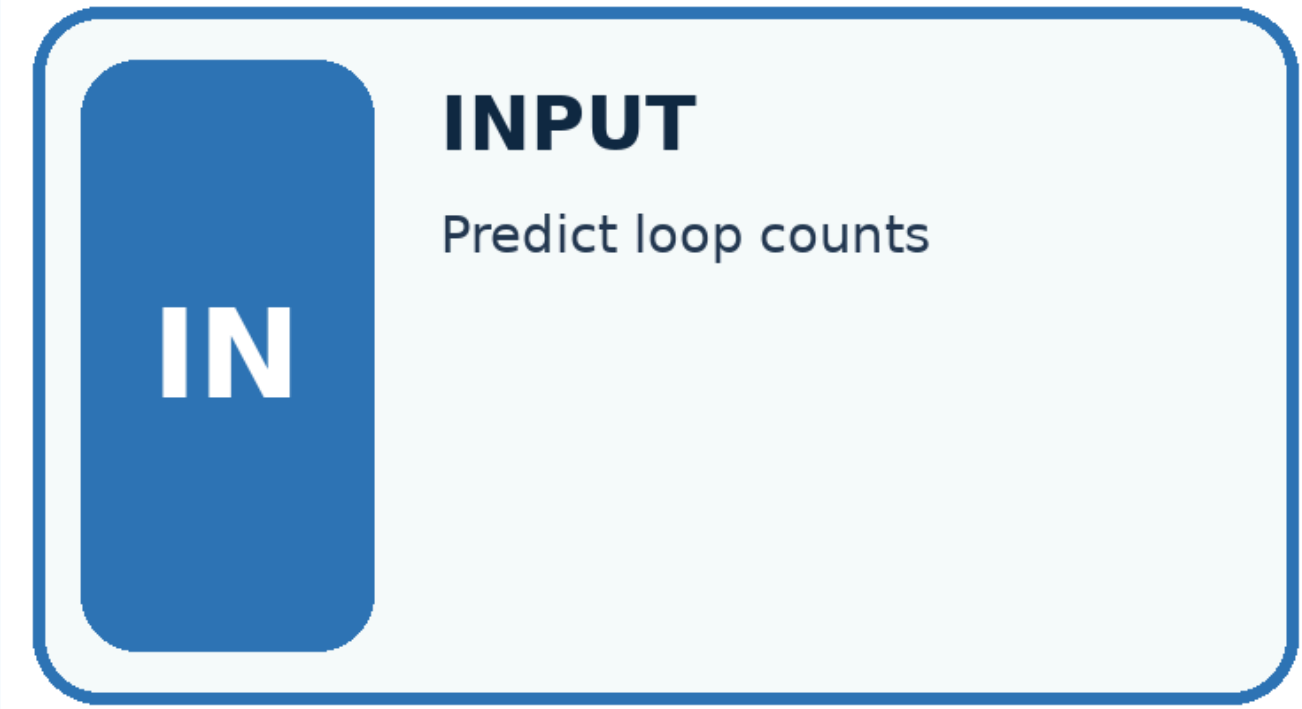
See how it works

> Follow the mechanism for Loops and Stopping. Connect each labelled input, deciding step and result.

VISUAL MECHANISM

Input → Process → Output

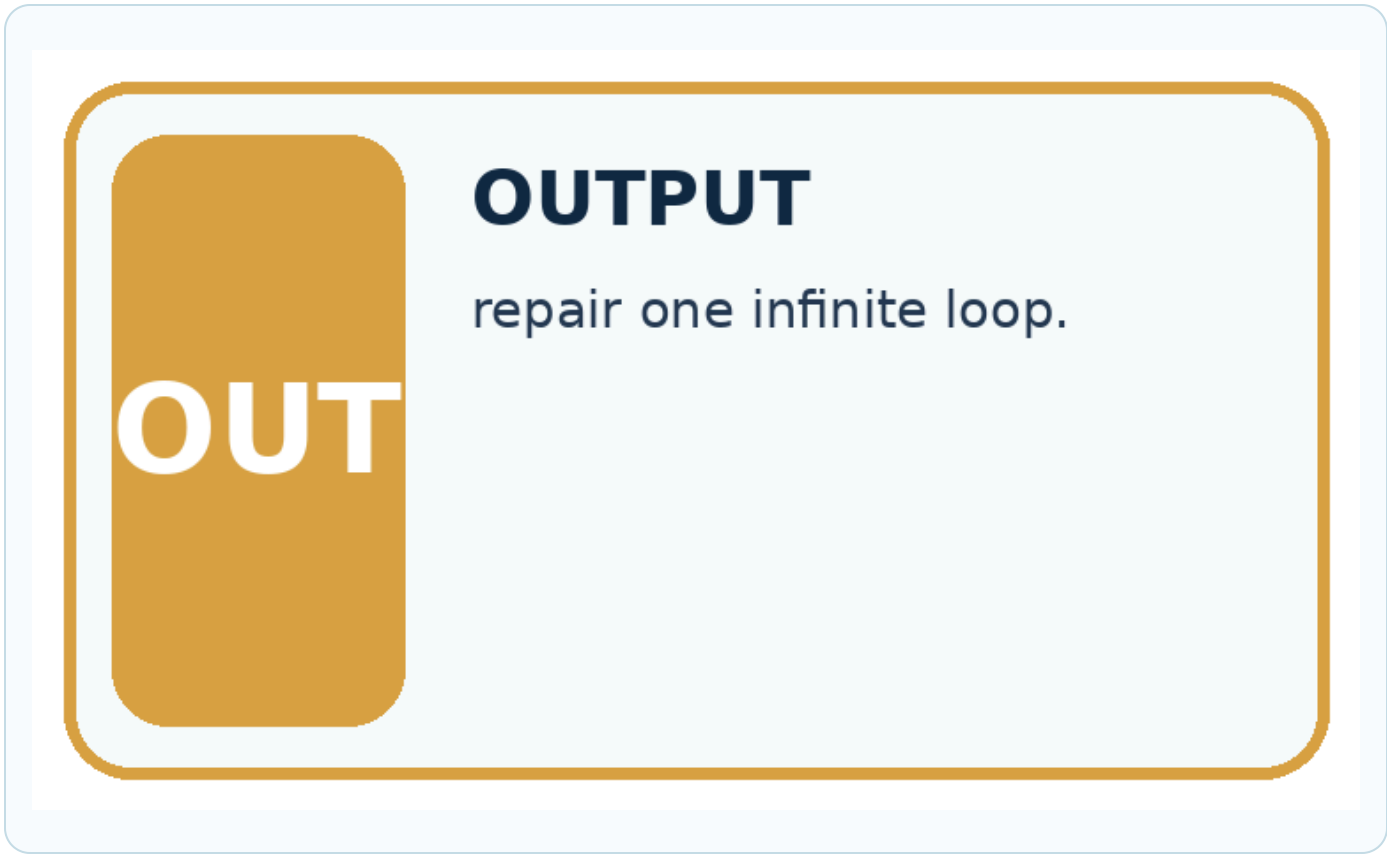
Trace the information. At each stage, ask what changed and how you could test it.



↓


↻
PROCESS
Useful repetition needs a clear update and stopping rule.

↓


OUT
OUTPUT
repair one infinite loop.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

```
total = 0
for turn in range(1, 4):
    total = total + turn
print(total)
```

State after each addition

TURN	TOTAL BEFORE	TOTAL AFTER
1	0	1
2	1	3
3	3	6

Initialize total to 0 before the loop. The unindented print runs once afterward.

The **output** is 6. Before the first **iteration**, total is 0. Each assignment reads the previous total and adds the current turn. After turn 3, the range is exhausted, so execution continues after the **loop**.

The table records state; it does not create extra instructions.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

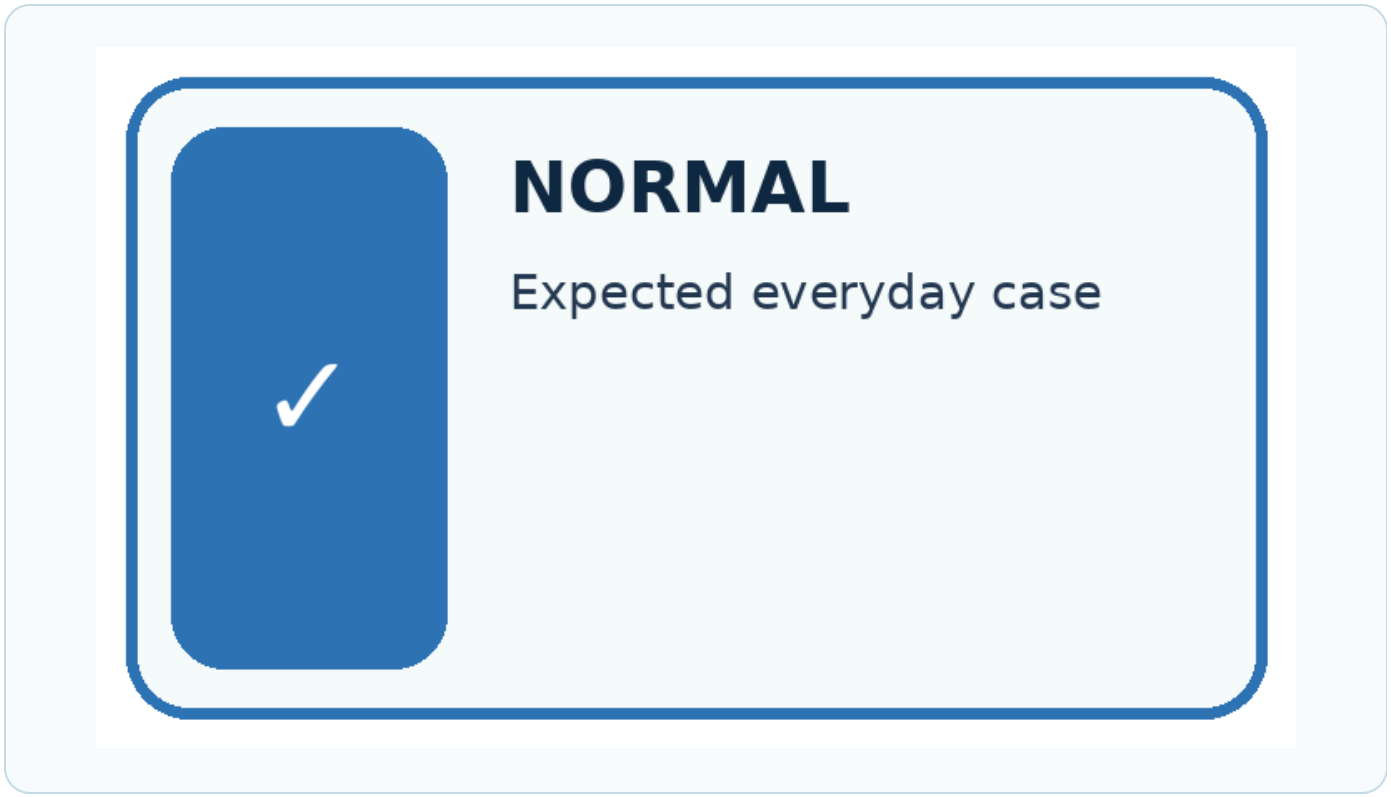
> Compare the supplied cases for Loops and Stopping. Identify the detail that changes the conclusion.

The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

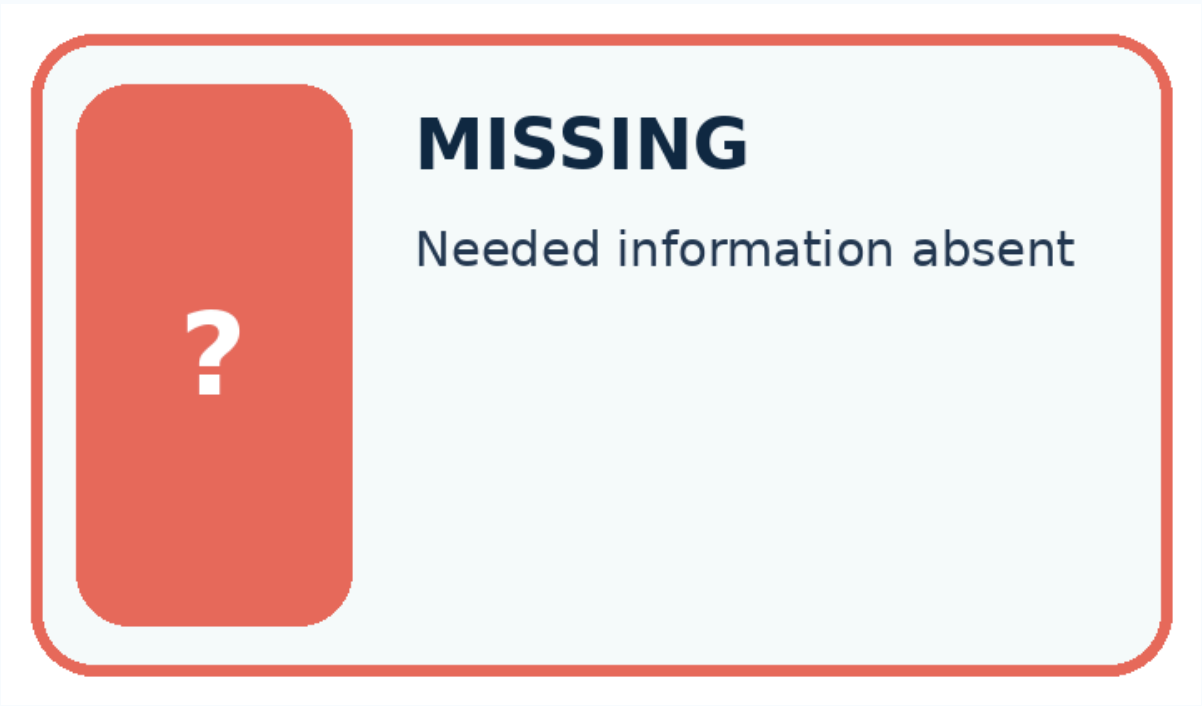
A system is not understood until it survives more than one comfortable example.


NORMAL
Expected everyday case


BOUNDARY
Near a decision edge

The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: Replace count>3 with count!=3.

Predicted result

Prediction: From 12, the **loop** never reaches 3 and does not terminate in this toy model.

Why the result follows

Reason: 12,8,4,0,−4,… are multiples of 4; none equals 3.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Loops and Stopping to explain a concrete decision, rather than only memorising definitions.

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

1

LOOP

instructions that repeat

2

UPDATE

a change made each repetition





Word	My own example	Not this / boundary
loop	A structure that repeats instructions according to a count or condition.	
update	Change a stored value or state according to a specified operation or new evidence.	
stopping rule	A specified condition that ends a repeated process.	

Word	My own example	Not this / boundary
infinite loop	A loop that never reaches its stopping condition under the assumed execution model.	

Words used in the closer explanation

Word	Meaning
Iteration	One execution of a loop body
Accumulator	A variable that keeps a running result
Initialization	Giving a variable its starting value
Trace table	A record of variable values as instructions execute

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STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Loops and Stopping, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 5 • WORKED EVIDENCE

1. Start count = 0.
2. Repeat while count < 4: show a card; count = count + 1.
3. The loop runs 4 times and stops when count becomes 4.
4. If count never changes, the stopping condition never becomes false.

1. Start count = 0.

2. Repeat while count < 4: show a card; count = count + 1.

3. The **loop** runs 4 times and stops when count becomes 4.

4. If count never changes, the stopping condition never becomes false.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Range expression	Values supplied	Number of iterations
range(1, 4)	1, 2, 3	3
range(1, 2)	1	1
range(1, 1)	No values	0
range(4)	0, 1, 2, 3	4

Worked explanation · Loops and Stopping

With one argument, range starts at zero. range(4) supplies four values but none is 4. If the worked script uses range(1, 1), the body never runs and the initialized total remains 0.

Do not rely on a variable first assigned inside a **loop**: a zero-**iteration** run might never assign it.

Replace range(1, 4) with range(1, 3) in the worked script. Predict the supplied values and final total. What stop value is needed to include 3?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Loops and Stopping

A fictional counter starts at 12 and decreases while above a boundary.

The facts we will use

Supplied facts

Python: count=12; while count>3: count=count-4; print after the **loop**.

Supplied facts

The condition is checked before each subtraction.

AI ACADEMY / AI-2/5

Loops and Stopping

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Initial check

12>3: true; subtract 4 to get 8.



2. Next check

8>3: true; subtract 4 to get 4.



3. Final update

4>3: true; subtract 4 to get 0.



4. Stop

0>3 is false; report 0 after three updates.

CHANGE ONE THING

Replace count>3 with count!=3.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Initial check	12>3: true; subtract 4 to get 8.
2. Next check	8>3: true; subtract 4 to get 4.
3. Final update	4>3: true; subtract 4 to get 0.
4. Stop	0>3 is false; report 0 after three updates.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

The condition does not promise the counter ends exactly at 3. Subtractions can cross the boundary. The count strictly decreases, so the below-or-equal stopping region is eventually reached from this start.

A condition requiring exact equality can be unsafe when repeated **update**s skip that value.

What this example does not establish

Keep the stated limit

Do not run an intentionally unbounded **loop** in class; trace a finite prefix to inspect the

defect safely.

Keep the stated limit

Read the labelled picture: Why is report 0 correct for the original **loop**, and why is equality-only stopping risky here?

Keep the stated limit

Change one thing: Replace count>3 with count!=3. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Loops and Stopping claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: Every repeated action will eventually stop by itself. Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Check the claim against the facts

Write the expected range values and **trace table** before running. Keep predicted and observed **output** separate, then explain the first row where they differ.

For a **loop** that should add a fixed amount repeatedly, zero **iteration**s should preserve the initial amount. One **iteration** checks one **update**; several check accumulation.

A learner checks the reset bug using only range(1, 2) and sees the expected answer 1. Explain why this does not establish correctness, and give a revealing test with its expected correct output.

STUDENT LESSON · 8 OF 18

Practise together

> Practise Loops and Stopping with support, then explain the deciding step in your own words.

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.

TRACE THE LOGIC

Predict the result. Trace every condition or calculation. Compare expected with actual. Explain the first difference.

SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

> Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

```
total = 0
for turn in range(1, 4):
    total = 0
    total = total + turn
print(total)
```

This faulty version prints 3. The extra reset discards earlier additions on every **iteration**. Remove the indented reset while keeping **initialization** before the **loop**.

Then test zero, one and several **iteration**s: the one-**iteration** case alone cannot reveal this reset bug.

Moving print(total) into the correct **loop** produces three lines, 1 then 3 then 6. Moving it out produces only the final 6. Trace both values and **output** events.

A **loop** follows written instructions; repeating them does not make the program learn.

Practise with less help: Trace before running

Use workbook W2 for the connected example "**Loop**s and **trace table**s". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Loop**s and **trace table**s. Proposed explanation: "The stop value is included". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Why is report 0 correct for the original **loop**, and why is equality-only stopping risky here?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

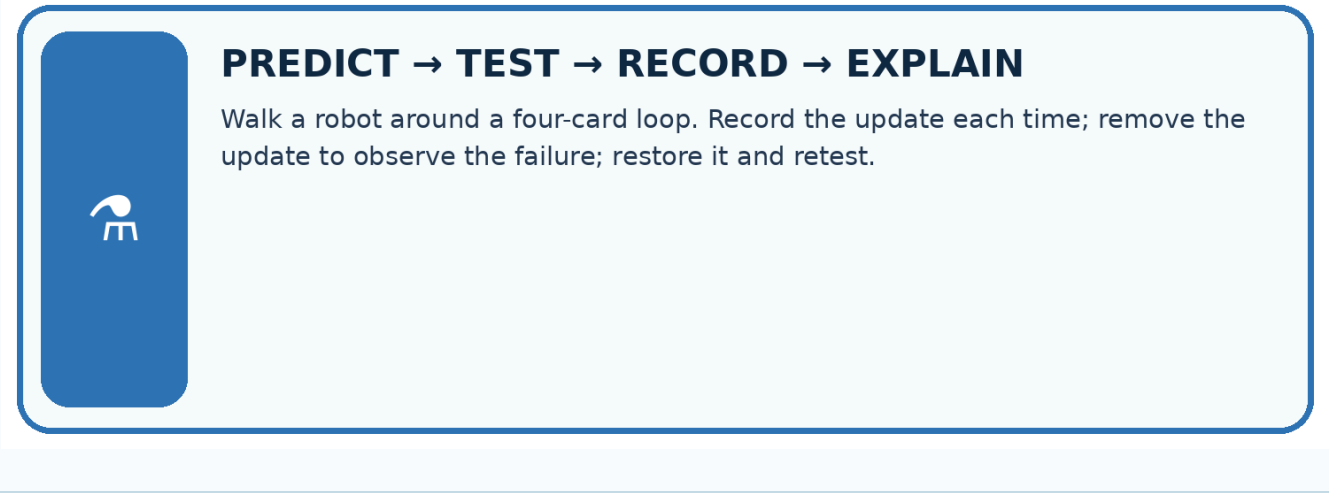
Run the investigation

> Investigate Loops and Stopping. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Walk a robot around a four-card **loop**. Record the **update** each time; remove the **update** to observe the failure; restore it and retest.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** boundary
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** missing
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** fresh transfer
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

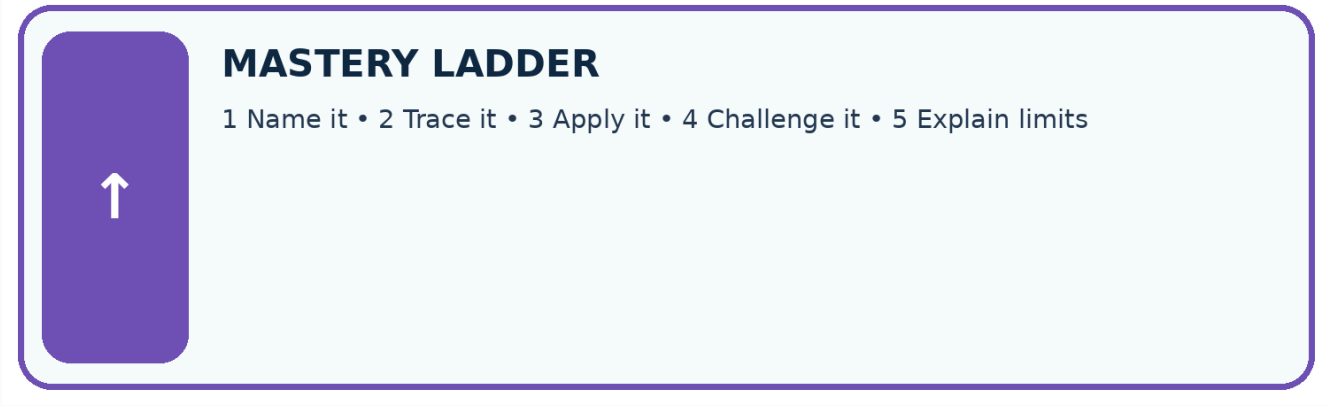
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Loops and Stopping. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

Write or draw your reasoning and evidence.

STUDENT LESSON · 11 OF 18

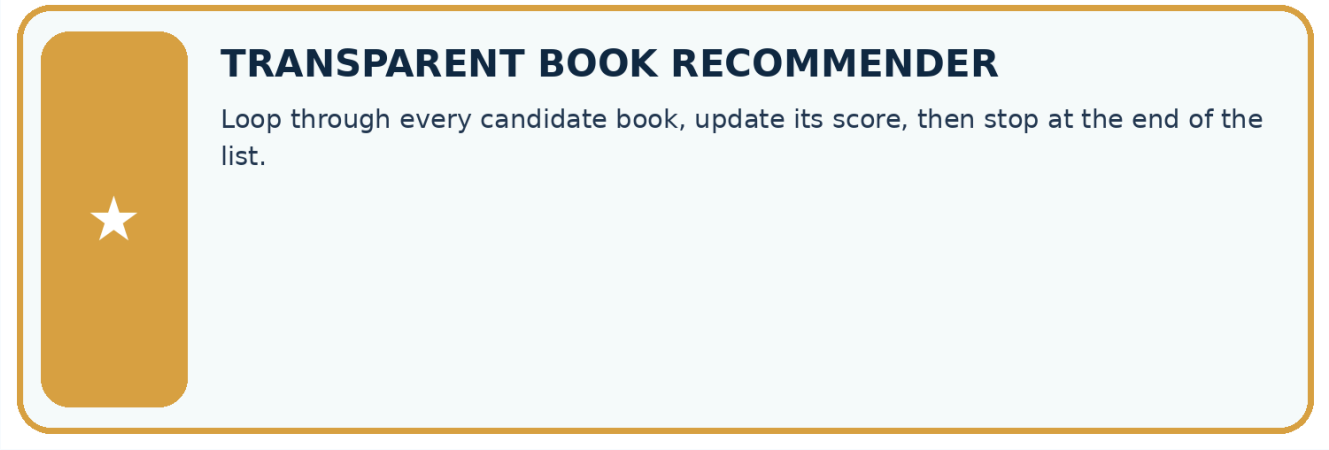
Connect your project

> Apply this week's idea to Transparent Book Recommender. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT **Loop** through every candidate book, **update** its score, then stop at the end of the list.

Trace the system

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Use progress and stopping arguments for any repeated recommender **process**; test **input**s that require zero, one and several **update**s.

STUDENT LESSON · 12 OF 18

Show what you know

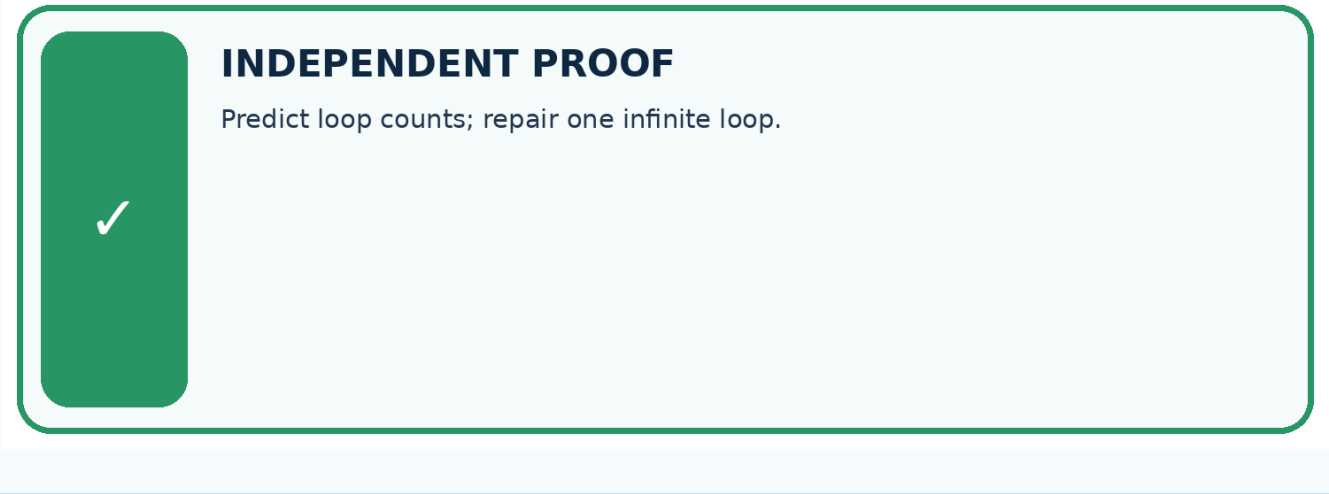
Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



Try a more demanding case

CHALLENGE Predict **loop** counts; repair one **infinite loop**.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Loops and Stopping case for supported practice. Repeated examples do not become fresh assessments.

Week 5 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Start n=0; while n<3, add 1 and record n.

The question

List records and stopping condition.

Worked explanation

1,2,3. The next comparison is false. The **update** makes progress toward termination.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

Trace the supplied case visually

A loop must progress toward stopping

Evidence or stage	Value or result
Start n	0
First update	1
Second update	2
Third update	3
Next n < 3 check	False, stop

Start n=0; while n<3, add 1 and record n.

List records and stopping condition.

1,2,3. The next comparison is false. The **update** makes progress toward termination.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Changed independent case

> Start count at 1. While count < 8, add 3. Trace the condition and values. What happens if the condition becomes count != 8?

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 5 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Start at count 0. While count is less than 3, add 1 and display count. List the displays and the stopping value.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Loop through every candidate book, **update** its score, then stop at the end of the list.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

> Trace count=0; while count<4: count=count+2. List values tested by the condition and final count. Would replacing <4 with !=5 terminate?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Loops and Stopping, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional counter starts at 12 and decreases while above a boundary.

By the end, I can	How I will show it
Trace variable values before and after each loop iteration	Explain the deciding step, show a trace or test, and state one limit.
Use the excluded stop value in range correctly	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Explain initialization and test zero one and several iterations	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Loops and Stopping without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Trace variable values before and after each **loop iteration**

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional counter starts at 12 and decreases while above a boundary.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? 0>3 is false; report 0 after three **update**s.

1. Next check

2. Final **update**

3. Stop

4. Initial check

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Stop 0>3 is false; report 0 after three **update**s. The condition does not promise the counter ends exactly at 3. Subtractions can cross the boundary.

The count strictly decreases, so the below-or-equal stopping region is eventually reached from this start. A condition requiring exact equality can be unsafe when repeated **update**s skip that value.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Replace count>3 with count!=3. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. From 12, the **loop** never reaches 3 and does not terminate in this toy model.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: From 12, the **loop** never reaches 3 and does not terminate in this toy model. 12,8,4,0,−4,… are multiples of 4; none equals 3.

Repair a missing building block

Review **Trace Table**s (ai-2/4). Goal: Trace variable values before and after each **loop iteration**. Short practice: Move print inside the original **loop**. What sequence is displayed?.

Return to this week and explain what you changed.

Review Decompose a Goal (ai-2/2). Goal: Distinguish a function definition from a function call. Short practice: What happens visibly if the caller stores needed but omits print(needed)?.

Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

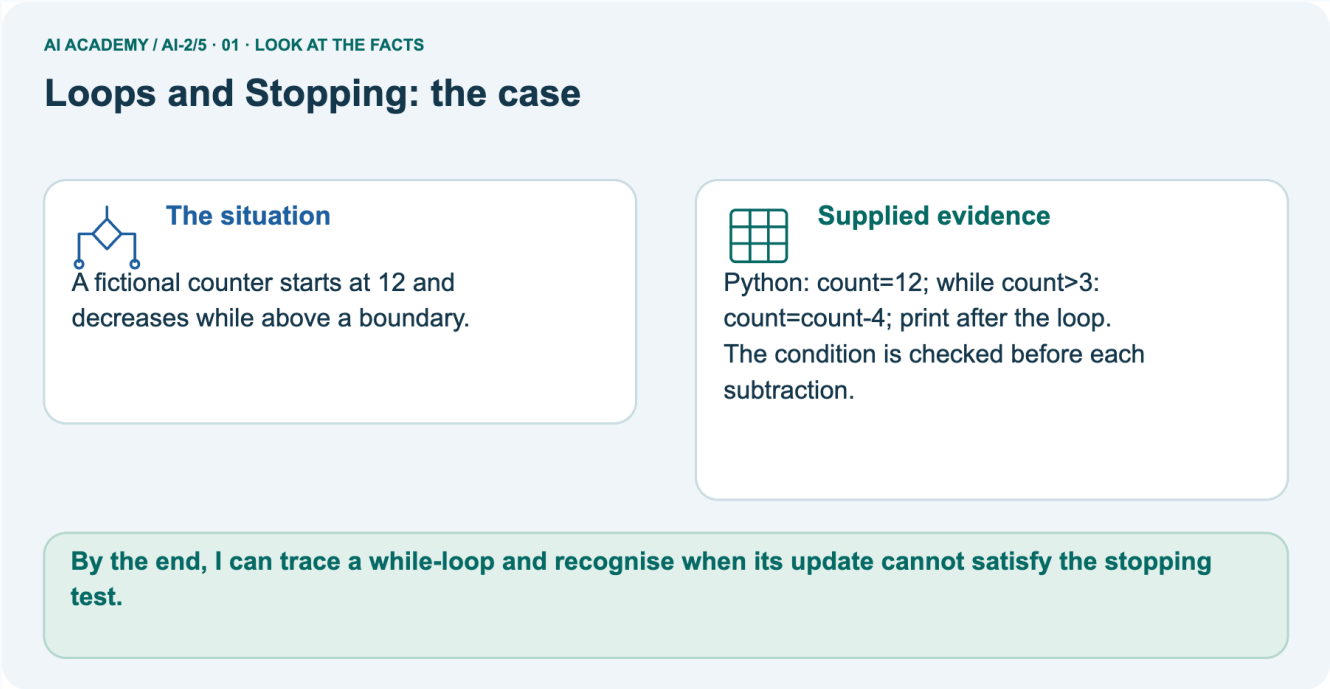
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Loops and Stopping graphic using its labels, deciding evidence and relevant limitation.

01. Loops and Stopping: the case

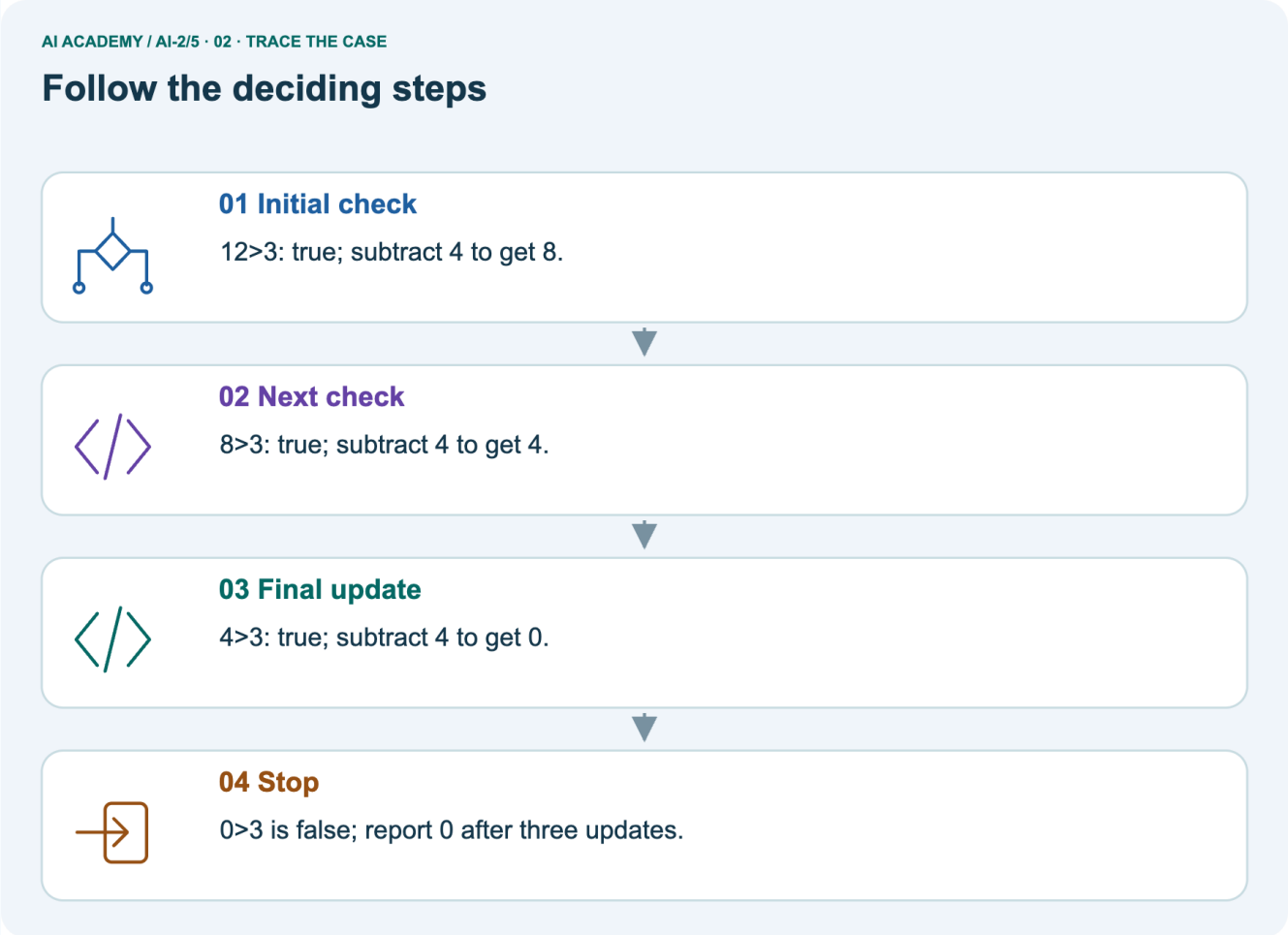


Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional counter starts at 12 and decreases while above a boundary. Supplied facts: Python: count=12; while count>3: count=count-4; print after the **loop**. The condition is checked before each subtraction.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Initial check: 12>3: true; subtract 4 to get 8. → Next check: 8>3: true; subtract 4 to get 4. → Final **update**: 4>3: true; subtract 4 to get 0. → Stop: 0>3 is false; report 0 after three **update**s.

Point to each numbered stage and explain which recorded **input** it uses.

03. State after each addition

AI ACADEMY / AI-2/5 · 03 · INSPECT THE LESSON MODEL

State after each addition

turn	total before	total after
1	0	1
2	1	3
3	3	6

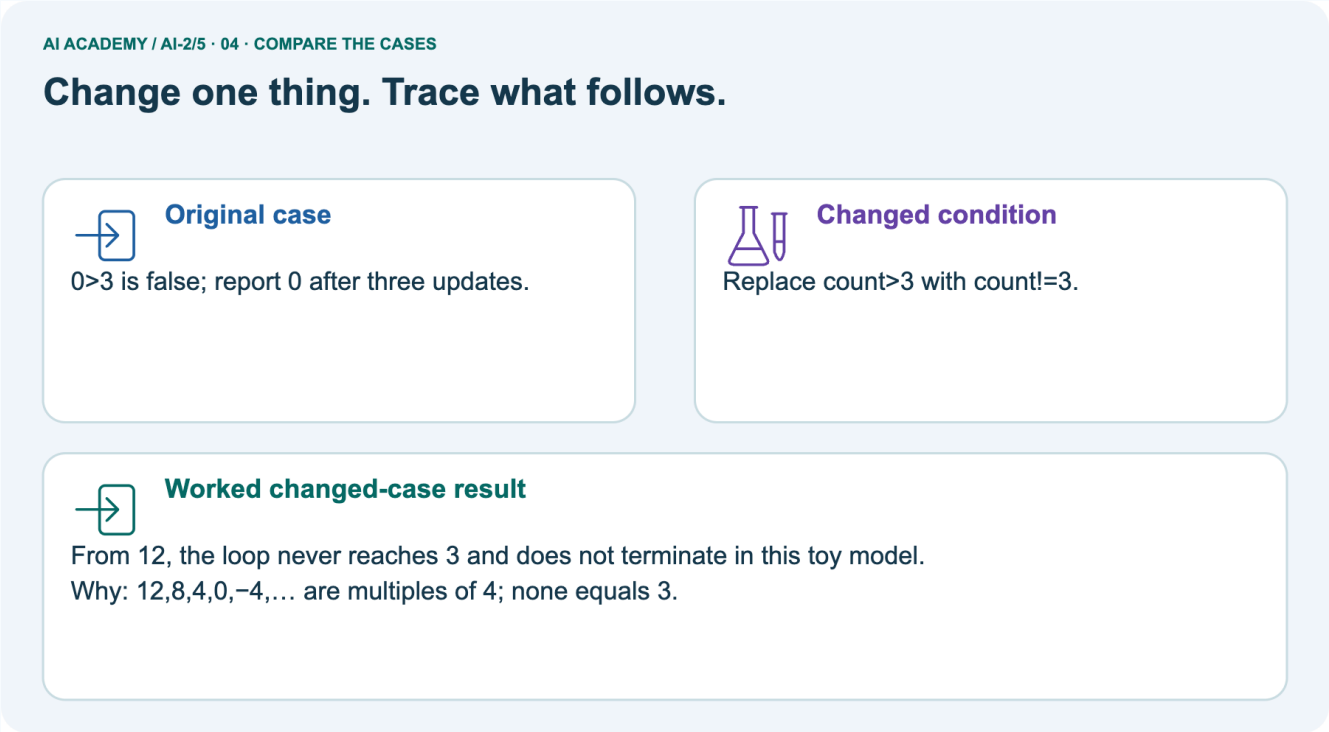
Initialize total to 0 before the loop. The unindented print runs once afterward.

Follow the labelled picture with the matching explanation. Open the vector diagram

State after each addition. 1 | 0 | 1. 2 | 1 | 3. 3 | 3 | 6

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



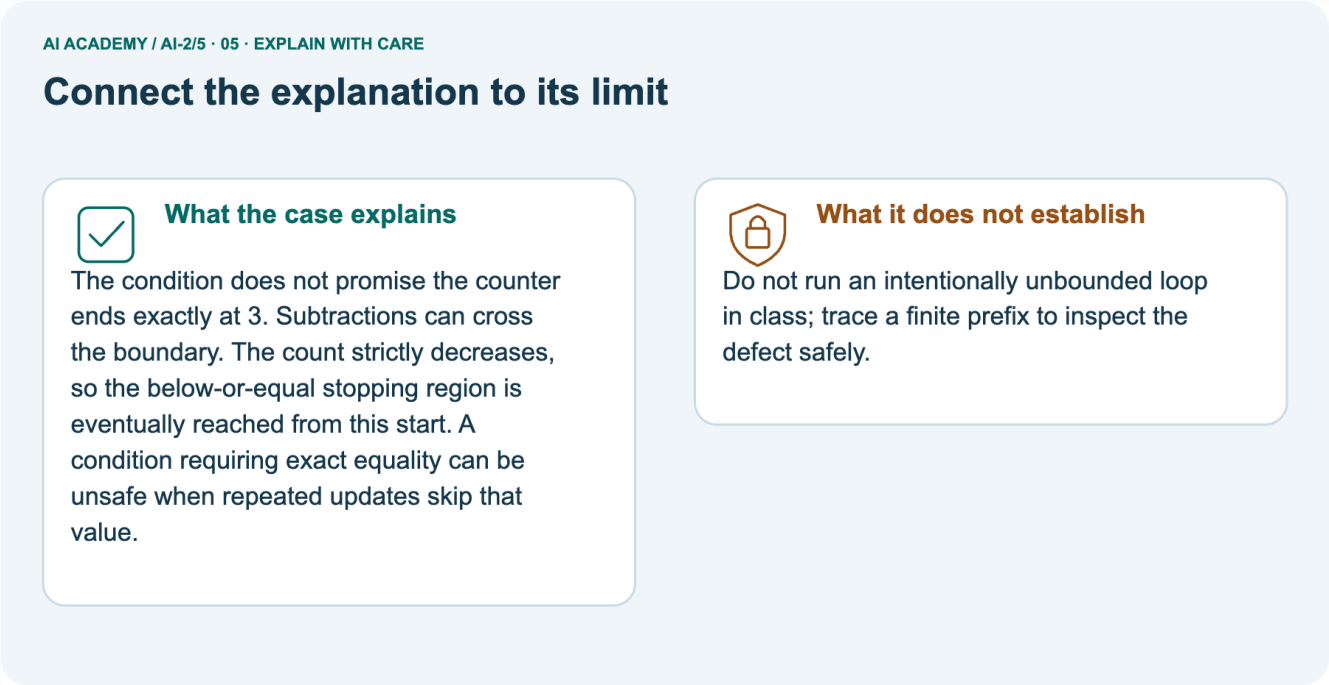
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: 0>3 is false; report 0 after three **update**s. Changed condition: Replace count>3 with count!=3. Expected result: From 12, the **loop** never reaches 3 and does not terminate in this toy model.

Reason: 12,8,4,0,−4,… are multiples of 4; none equals 3.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



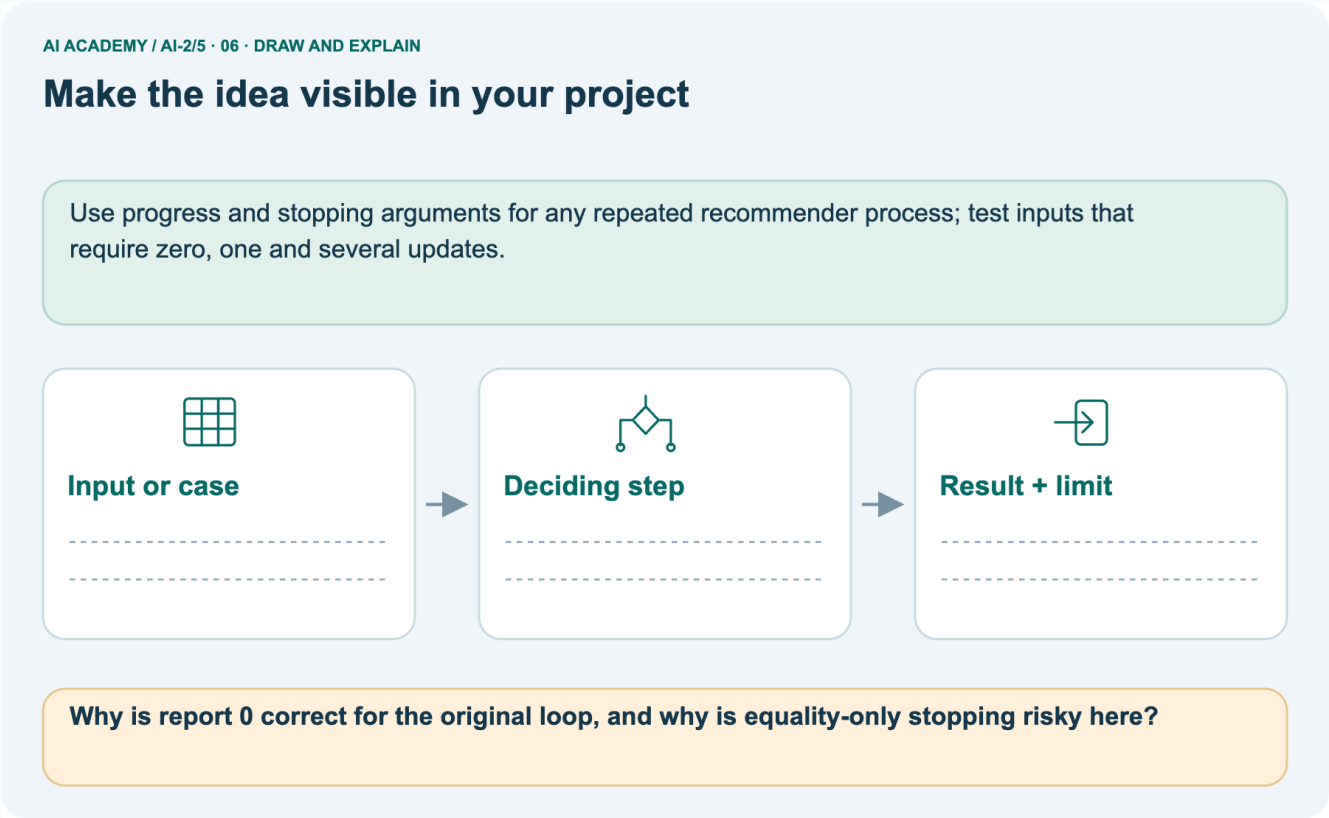
Follow the labelled picture with the matching explanation. Open the vector diagram

The condition does not promise the counter ends exactly at 3. Subtractions can cross the boundary. The count strictly decreases, so the below-or-equal stopping region is eventually reached from this start.

A condition requiring exact equality can be unsafe when repeated **update**s skip that value. Limit: Do not run an intentionally unbounded **loop** in class; trace a finite prefix to inspect the defect safely.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Use progress and stopping arguments for any repeated recommender **process**; test **input**s that require zero, one and several **update**s. Draw your **input**, deciding step, result and limit.

Supported question: Why is report 0 correct for the original **loop**, and why is equality-only stopping risky here?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.